Seeing Beyond the Black Box: Explainable AI for Handwriting Author Identification

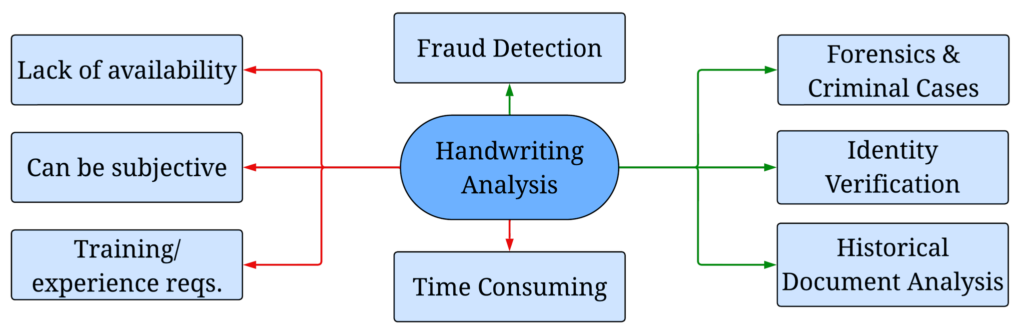
THE UNIVERSITY OF TEXAS AT ARLINGTON

Zainab Siddiqui, zxs2546@mavs.uta.edu
Division of Data Science, College of Science,
The University of Texas at Arlington

csafe
Advancing Forensic Practice Through Statistics and AI

Background

- Handwriting authorship analysis applications & limitations:



- Can computer vision models make this more efficient & objective?
- Can their decisions be trusted and their predictions be held accountable for real-world, consequential applications?

Goal: Investigating the feasibility and trustability of predicting author identity based on handwriting analysis using convolutional neural networks (CNNs).

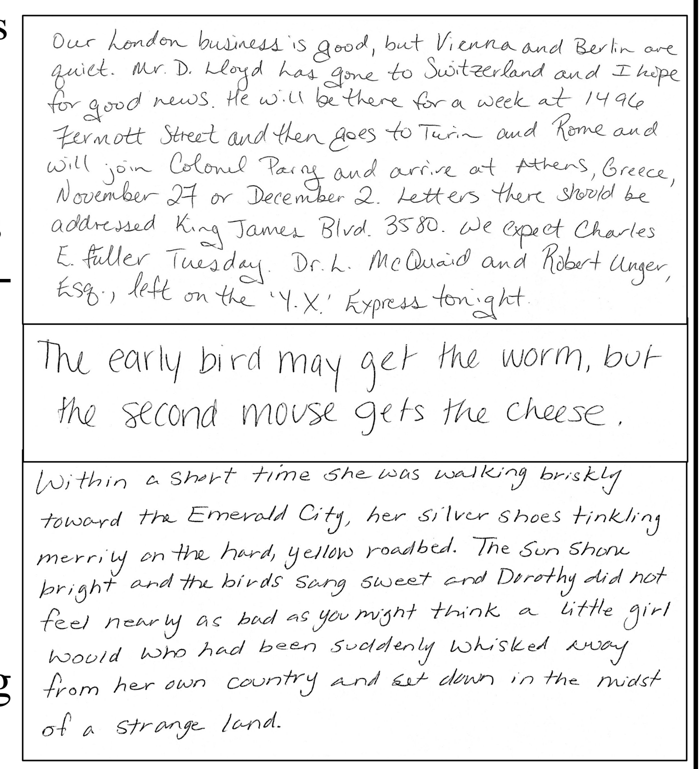


Figure 1: Example data images.

Methods

- Transfer learning CNN models with TensorFlow & Keras
- Special features of MobileNetV2: light-weight, inverse residual bottlenecking

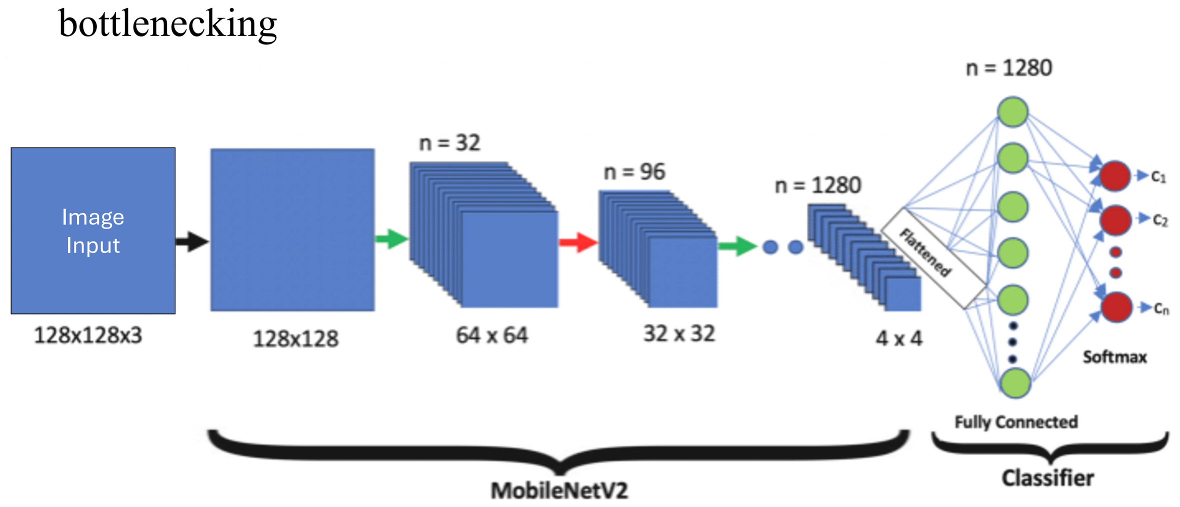


Figure 2: MobileNetV2 general architecture (Akay et al.)

- Computational GPU resources from Google Colab Pro (A100 GPU)
- Images padded or cut to squares and downsized; expanded grayscale images to 3-channel RGB
- Model training hyperparameters:

Epochs	Batch Size	Image Size	Classes	LR
80–100	64	384×384×3	475	Adam (1e-3)

- 27 images per class = 12,825 samples → 15/6/6 train/val./test split
- Main metric: top-3 accuracy; more forgiving than raw accuracy
- XAI interpretation through Grad-CAM & LIME

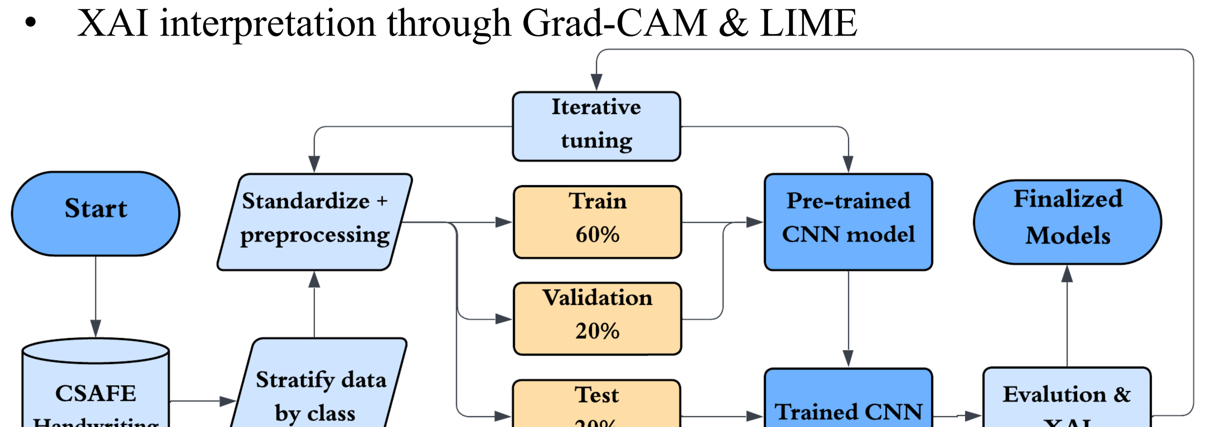


Figure 3: Modelling pipeline overview flowchart.

Results

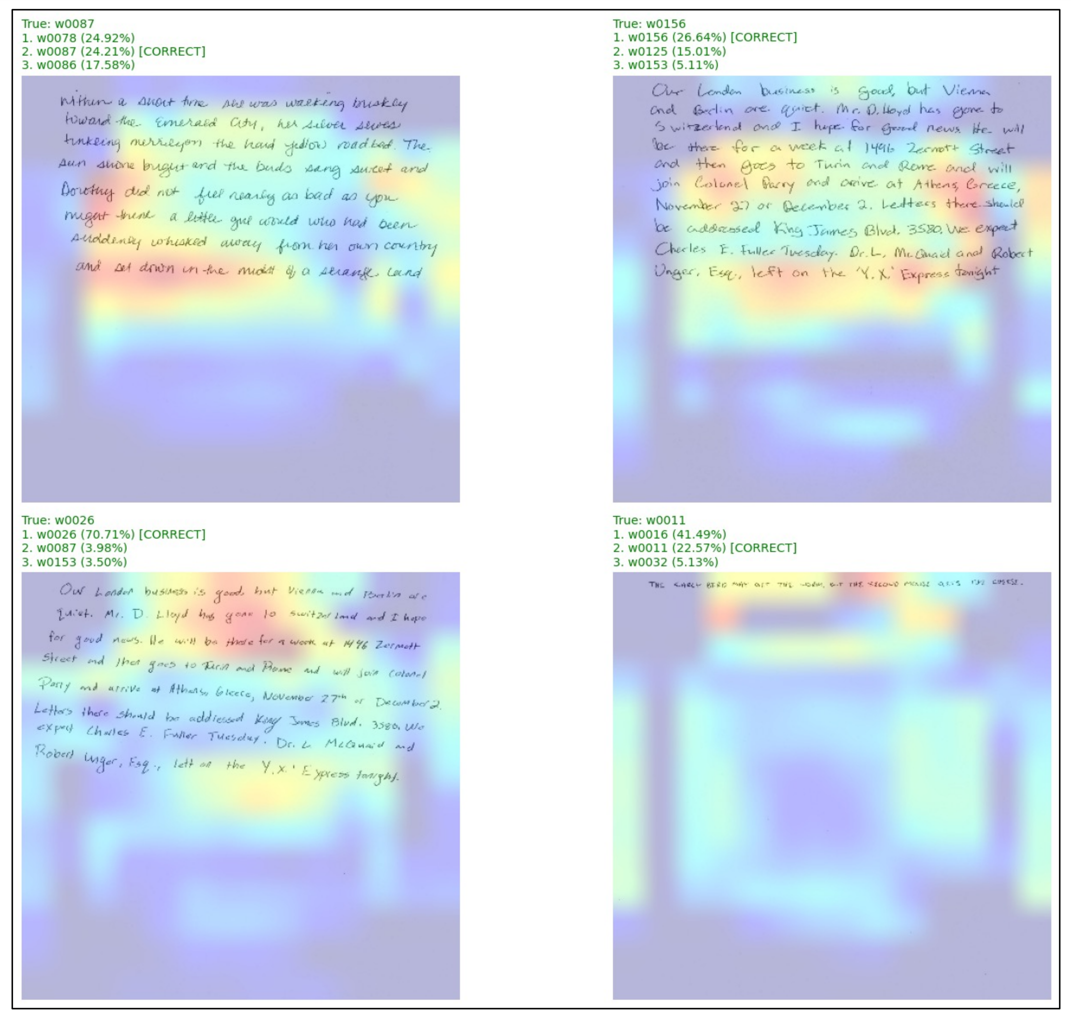


Figure 4: Grad-CAM on earlier model, trained on a 90-class subset, showing more consistent and "proper" gradients of model focus.

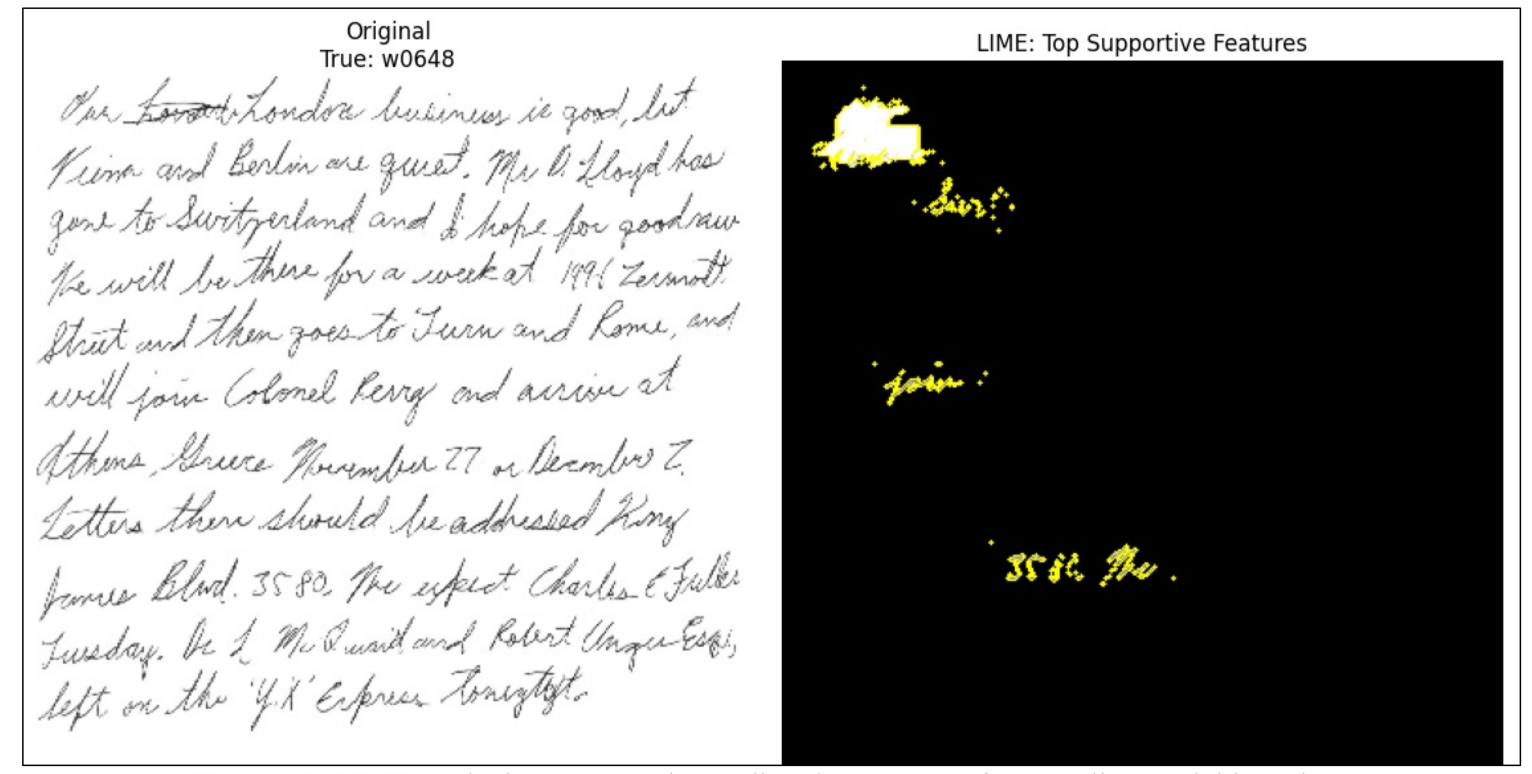


Figure 5: LIME analysis on correctly predicted test output from earlier model iteration, highlighting specific words.

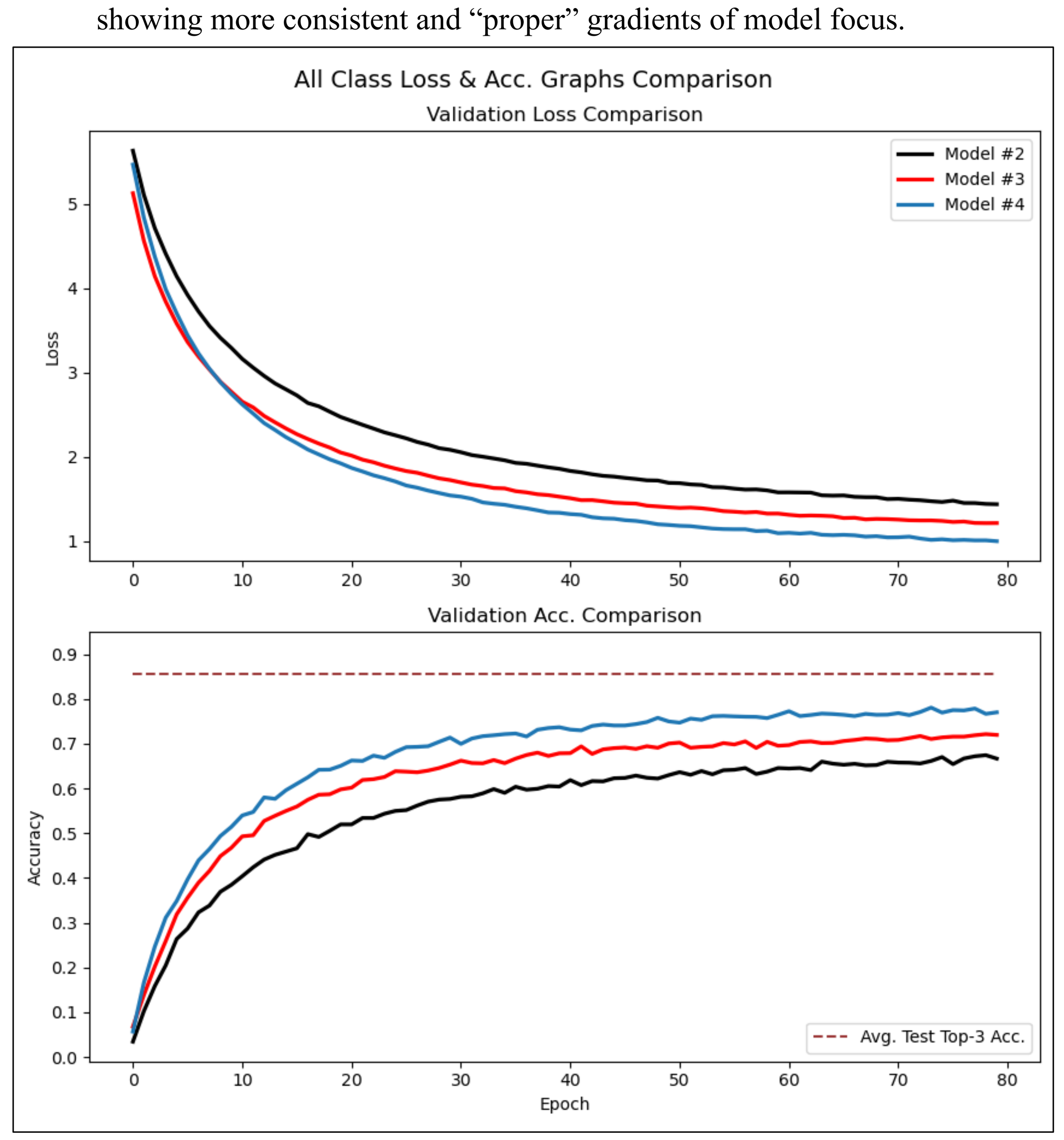


Figure 6: Model performance comparison showing performance plateauing at approximately 70% accuracy and average top-3 test accuracy.

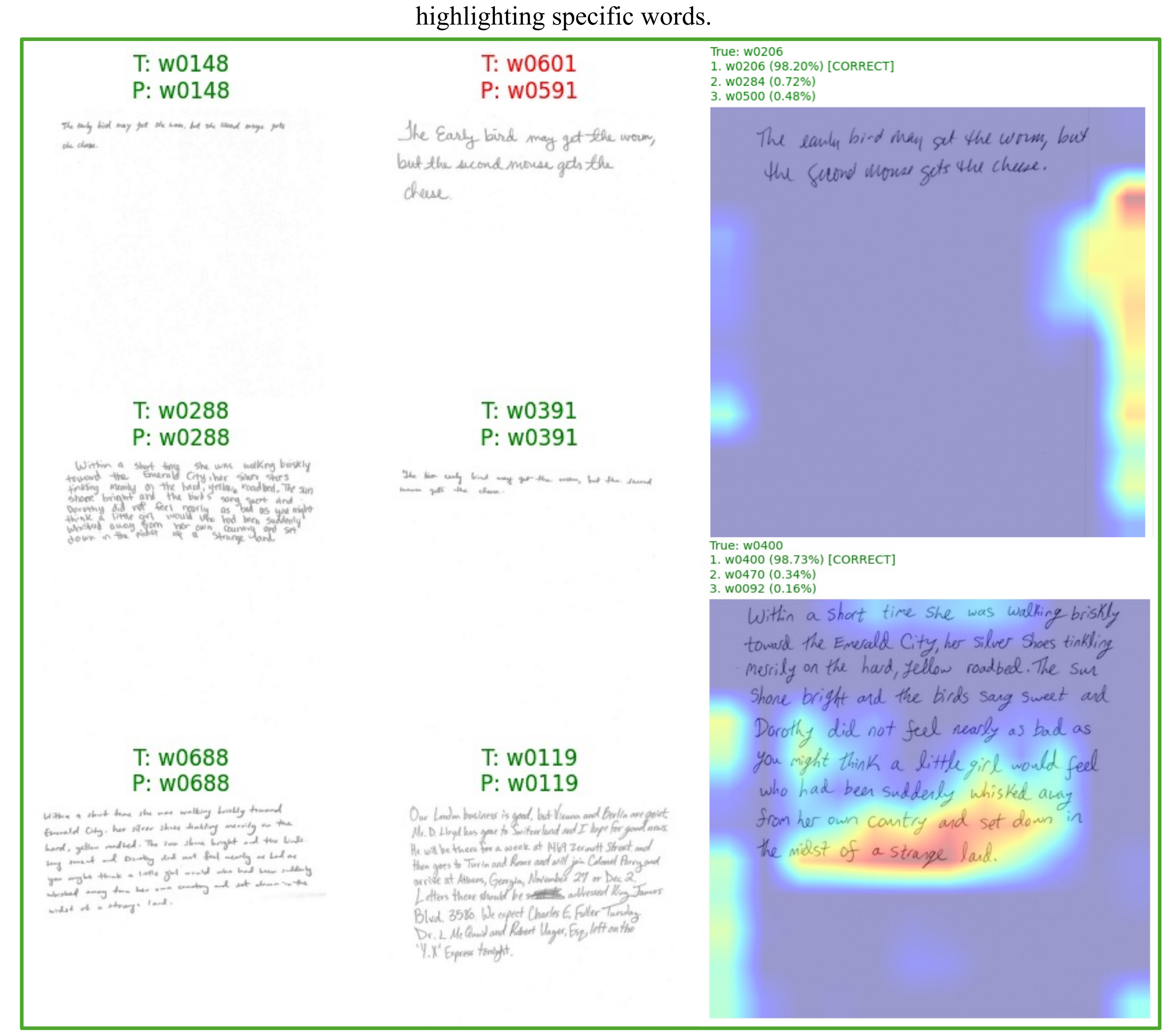


Figure 7: Testing output from Model #4 showing correct and incorrect predictions (left). Grad-CAM heatmaps highlighting regions of significance and showing top-3 predictions (right).

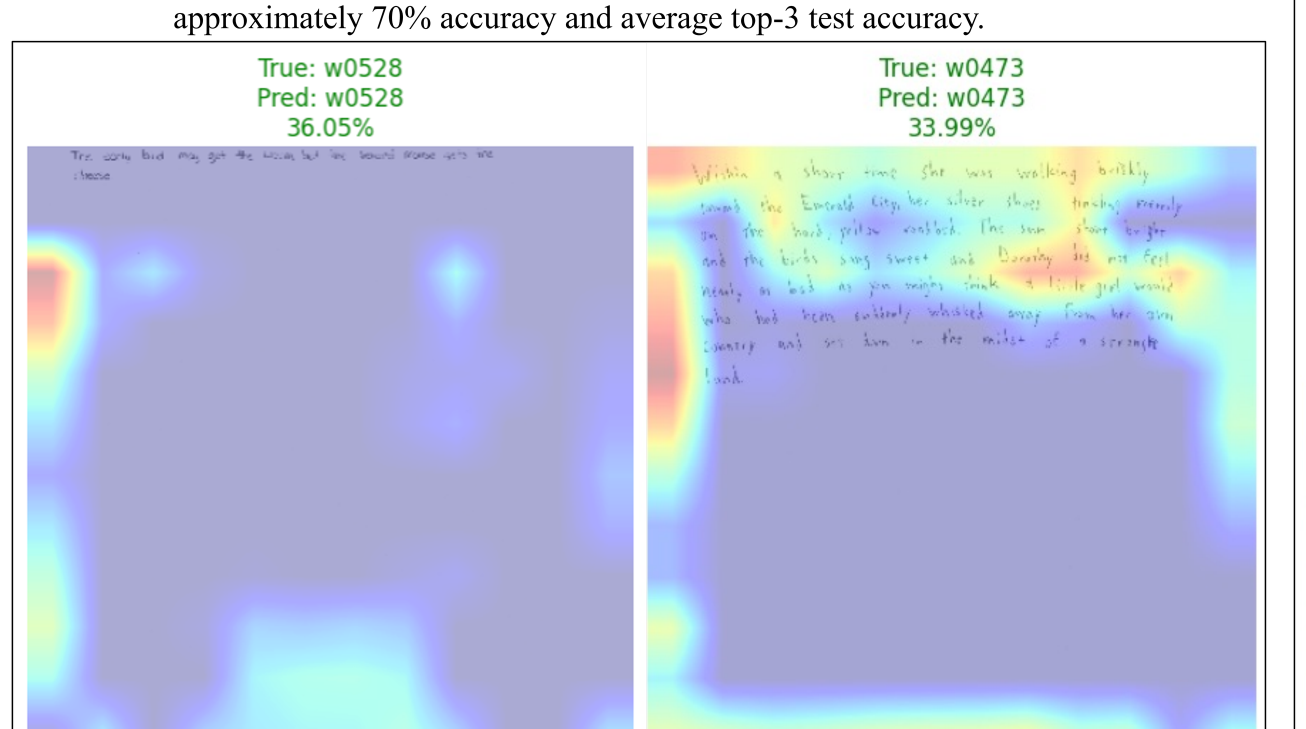


Figure 8: Grad-CAM analysis of testing outputs from Model #2, highlighting regions of significance and showing correct predictions, despite "incorrect" focus.

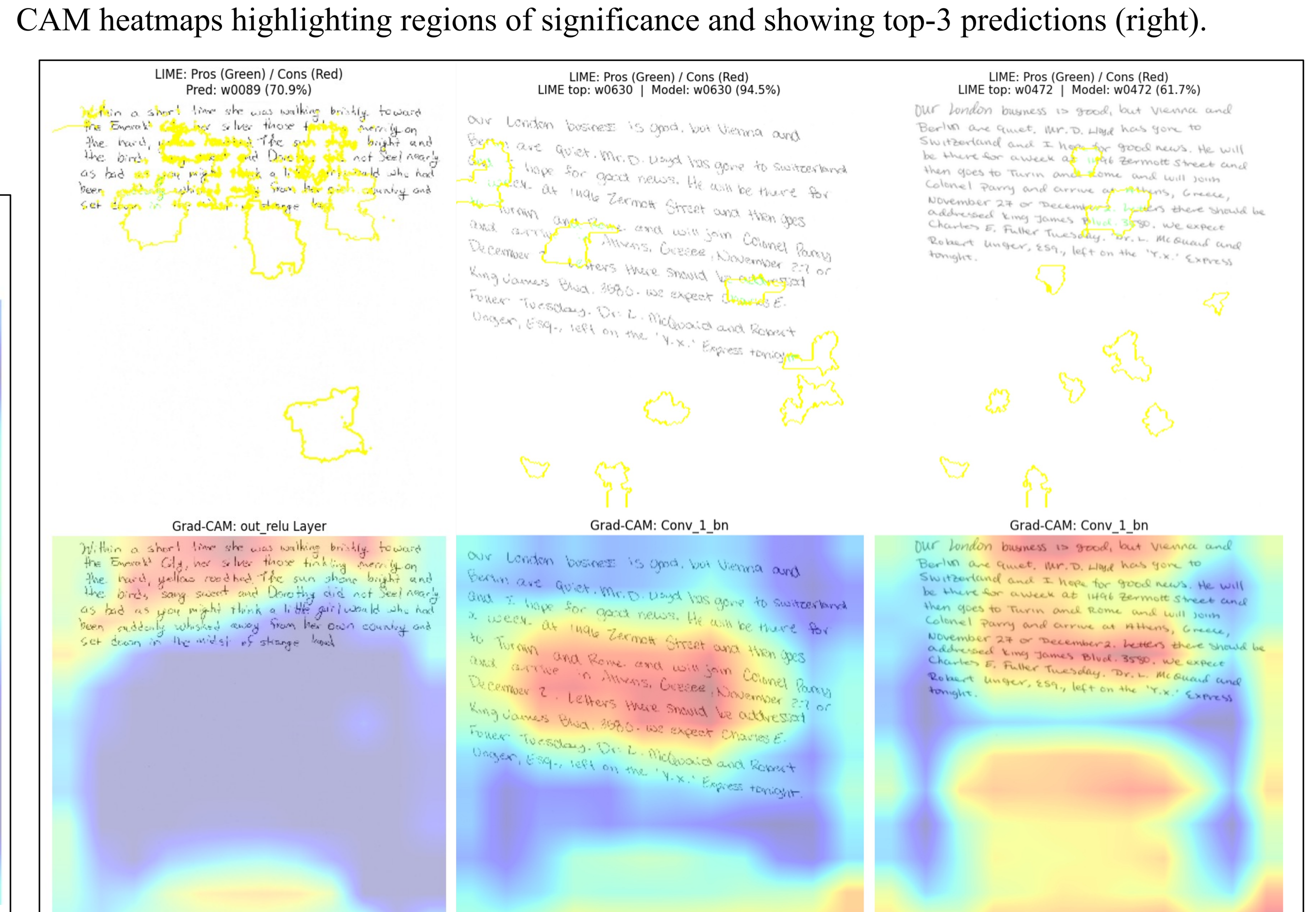


Figure 9: LIME + Grad-CAM analysis on test outputs from (left to right) Model #3, #2, and #4.

Discussion

- Handwriting features *do* appear to be a viable predictor of author identity
- Minor overfitting present (i.e., train acc. > test acc.), mitigated through top-3 accuracy
- Grad-CAM shows model overreliance on margins and whitespace
  - Does not appear to correlate with padded images/areas
  - Indicates some predictions lack proper justification
- Both LIME & Grad-CAM results seem to degrade/become more random in models with more cleaning/preprocessing
- Limitations:
  - Limited generalizability; i.e., closed set model
  - Preprocessing (i.e., normalizations, augmentations) administered to test set to standardize inputs
- "Best" model is difficult choose; numbers vs. explainability

Table 1: Metric comparison across most recent models.

Metric	Model #2 (Square)	Model #3 (Normal)	Model #4 (Square + Cut)
Train Acc.	87.8%	91.3%	**95.1%**
Val. Top-3 Acc.	82.7%	86.3%	**89.8%**
Test Acc.	67.3%	70.8%	**77.4%**
Test Top-3 Acc.	81.8%	85.6%	**89.6%**

Future work

- Black and white color grading normalization & natural grayscale (versus default RGB) handling
- Bounding boxes to focus on text
- More aggressive preprocessing
  - E.g., more extreme cropping & zooming, focusing on words/letters
- Heavier & larger transfer learning architectures
- Deployment demo showing conceptual feasibility

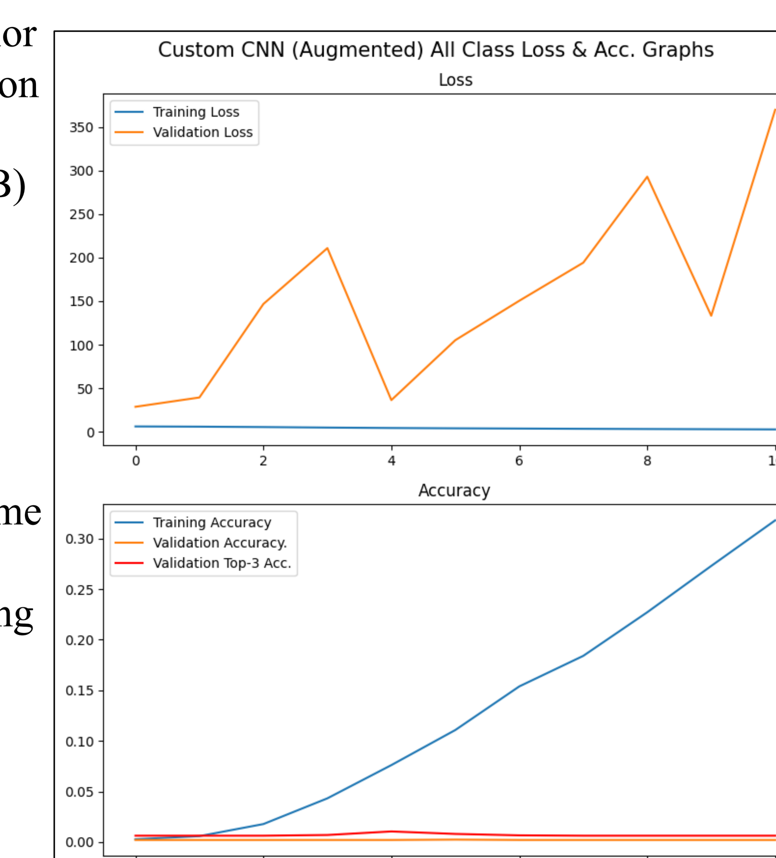


Figure 10: Custom CNN's loss & accuracy curves (early stopping after validation plateau), showing non-transfer learning models are unfit and infeasible for this problem.

Citations

Akay, Metin & Du, Yong & Sershen, Cheryl & Wu, Minghua & Chen, Ting & Assassi, Shervin & Mohan, Chandra & Akay, Yasemin. (2021). Deep Learning Classification of Systemic Sclerosis Skin Using the MobileNetV2 Model. IEEE Open Journal of Engineering in Medicine and Biology. PP. 1-1. 10.1109/OJEMB.2021.3066097. (Image)

Crawford, A., Ray, A., Carriquiry, A., Kruse, J., & Peterson, M. (2019). CSAFE Handwriting Database. https://doi.org/10.25380/u002Fiastate.10062203.v1. Dataset.